

4th Summer School on ML for Electron Microscopy

University of Tennessee Knoxville, with Oak Ridge National Laboratory, Brookhaven National Laboratory, Thermo Fisher, AtomQ, Mat3ra

Organizers: Gerd Duscher, Sergei V. Kalinin

Tutorials by: Austin Houston, Boris Slautin, Elizabeth Heon, Sheryl Sanchez, Paolo Longo (TF), Kevin Roccapiore (AtomQ), Andrew Lupini (ORNL), Kamal Choudhary (JHU), Colin Ophus (Stanford)

If you are working on machine learning for microscopy - whether for data analysis, real-time analytics, or autonomous instrument control - keep your schedule open for the **Summer School on ML/AI for Electron Microscopy**, hosted by the University of Tennessee, Knoxville.

📍 **When?** June 22–26, 2026

📍 **Where?** Hybrid format (in-person + online)

This year, the school is explicitly structured around a **top-down and bottom-up integration of AI in microscopy**:

- **Top-down: Agentic AI for instrumentation** — how to design AI agents that interact with microscopes through APIs and orchestrators, enabling remote operation, cloudified workflows, and federated experimental systems.
- **Bottom-up: Reward-driven workflows** — how to build decision-making frameworks grounded in physics and experimental feedback, enabling robust, adaptive, and autonomous exploration.

The school will cover core ML concepts, but the primary focus is on **next-generation AI-enabled microscopy ecosystems**, including:

✅ **Agentic control of instruments** – connecting LLM-driven agents to microscopes via APIs and orchestration layers.

✅ **Reward-driven experimental design** – formalizing objectives and enabling autonomous decision-making beyond heuristic workflows.

✅ **Cloudification and federation** – transforming microscopes into distributed, connected platforms operating across sites.

✅ **From data to action** – closing the loop between acquisition, analysis, and decision-making in real time.

Participants will work with **realistic microscopy datasets and digital twins**, enabling development and testing of ML workflows under conditions that reflect actual experimental constraints. The emphasis is on transitioning from **human-controlled experiments** to **AI-augmented and ultimately agent-driven systems**, where decision-making happens at the time scale of the instrument.

Hands-on sessions and hackathon-style modules will ensure that participants engage directly with:

- **API-level control and orchestration of instruments**
- **Integration of ML models into live experimental workflows**
- **Stochastic optimization and decision-making under uncertainty**
- **Coupling between data streams, models, and actions**

Whether your interest is in microscopy data analysis, automated control, or building the next generation of **cloud-connected, AI-driven scientific instruments**, this school provides a concrete pathway from concepts to implementation.

#ElectronMicroscopy #MachineLearning #ArtificialIntelligence #AutonomousMicroscopy
#AgenticAI #CloudScience #ScientificAI #MaterialsScience

Monday, June 22: Atomic Resolution STEM and Physics from Atomic Positions

8:30 AM–9:00 AM Welcome and Introduction (Kalinin)
9:00 AM–10:00 AM Principles of Electron Optics and Aberration Correction in STEM (Duscher)
10:00 AM–10:50 AM Remote Aberration Correction in STEM (Duscher, Houston)
11:10 AM–12:00 PM Simulation of Ronchigrams [Participant computer] (Duscher)
12:00 PM–1:00 PM Lunch (provided for registered onsite participants)
1:00 PM–2:00 PM Cloudifying STEM: APIs, cybersecurity, operation (Duscher, Houston)
2:00 PM–2:50 PM Introduction to Neural Networks for Images (Heon)
3:10 PM–4:00 PM Physics from STEM data [Participant computer] (Kalinin, Austin)
4:00 PM–5:00 PM Hackathon 1: Image analysis pipelines

Tuesday, June 23: Electron Diffraction and 4D-STEM

9:00 AM–10:00 AM Introduction to Diffraction (Duscher)
10:00 AM–10:50 AM Remote Diffraction Acquisition in STEM (Duscher, Houston)
11:10 AM–12:00 PM Simulation of Diffraction Pattern (Duscher)
12:00 PM–1:00 PM Lunch (provided for registered onsite participants)
1:00 PM–2:00 PM Analyzing Diffraction pattern [Participant computer] (Duscher, Houston)
2:00 PM–2:50 PM (guest lecture) 4D STEM (Ophus, Stanford)
3:10 PM–4:00 PM Conventional and Smart Acquisition of 4D STEM (Houston)
4:00 PM–5:00 PM Hackathon 2: Data curation and unsupervised analysis of the diffraction data

Wednesday, June 24: Spectroscopic Data and ML-enabled analysis

9:00 AM–10:00 AM Introduction to Spectroscopy (Duscher)
10:00 AM–10:50 AM Remote Acquisition of EDS spectra in STEM (Duscher, Houston)
11:10 AM–12:00 PM Analysis of EDS [Participant computer] (Duscher)
12:00 PM–1:00 PM Lunch (provided for registered onsite participants)
1:00 PM–2:00 PM Machine Learning of Spectroscopic Datasets (Kalinin)
2:00 PM–2:50 PM VAE for Image and Spectrum Analysis (Kalinin)
3:10 PM–4:00 PM Conventional Analysis of EELS [Participant computer] (Duscher)
4:00 PM–5:00 PM Hackathon 3: ML-Enhanced Analysis of EELS

Thursday, June 25: ML for STEM: from post-acquisition to real time analytics

9:00 AM–10:00 AM Introduction to Workflows in Machine Learning (Kalinin)
10:00 AM–10:50 AM Building the STEM control agents (Sanchez)
11:10 AM–12:00 PM Federating STEM: from machine to network (Kalinin, BlueSky)
12:00 PM–1:00 PM Lunch (provided for registered onsite participants)
1:00 PM–2:00 PM (guest lecture) Autonomous Operation (Longo, TF)
2:00 PM–2:50 PM ML-based tuning and drift correction (Sgouralis)
3:10 PM–4:00 PM (guest lecture) Agentic analysis and streaming to theory (Choudhary, JHU)
4:00 PM–5:00 PM Hackathon 4: Building agents for STEM

Friday, June 26: Automated and Autonomous Electron Microscopy

9:00 AM–10:00 AM Decision Making in Microscopy (Kalinin)
10:00 AM–10:50 AM Reward Functions for Decision Making (Kalinin)
11:10 AM–12:00 PM Principles of Gaussian Processes and Bayesian Optimization (Kalinin)
12:00 PM–1:00 PM Lunch (provided for registered onsite participants)
1:00 PM–2:00 PM Deep Kernel Neural Learning and hAE (Kalinin)
2:00 PM–2:50 PM AE Microscopy for Materials Exploration (Slautin)
3:10 PM–4:00 PM (guest lecture) Atomic Fabrication with STEM (Roccapiore, AtomQ)
4:00 PM–4:30 PM (guest lecture) New opportunities for STEM (Lupini, ORNL)
4:30 PM–5:00 PM Conclusion (Kalinin and Duscher)